



Airbnb Candolim Stay — Production Scaling Architecture

System Architecture Blueprint: Frontend, Storage, Search, and Deployment Strategies



1. Frontend Scaling

Edge Delivery & Client Performance

Global Edge CDN & Caching

< 25ms TTFB

Anycast edge network (Cloudflare / CloudFront) caching static assets, JS/CSS bundles, and HTML at 300+ PoPs worldwide with stale-while-revalidate policies.

CloudflareBrotli/GzipHTTP/3

Dynamic Responsive Media Pipeline

85% Bandwidth Saved

Dynamic width parameterization (`im_w=720/1200`) with automatic WebP/AVIF transcoding and responsive srcset for 43+ high-res room photos.

getOptimizedImageUrlWebP / AVIFLazy Loading

Incremental Static Regeneration (ISR)

High Concurrency

Pre-rendered static HTML for top Candolim listings with background revalidation on price/calendar updates, offloading 98% of traffic from origin app servers.

Edge SSRMicro-cachingVite / Node

Client State & Accessibility

Zero Layout Shift

Modular architecture with `useLockBodyScroll` and `useFocusTrap`. Lazy-loaded Photo Tour & Lightbox dialogs with pure CSS sticky room headers.

React 19Pure CSS StickyWCAG 2.1 AA



2. Storage Scaling

Tiered Data, Caching & Object Stores

PostgreSQL Read/Write Split

ACID Compliance

Primary instance handles atomic bookings, calendar blocks, and price mutations. Scalable read-replica cluster behind PgBouncer handles listing queries.

Aurora PostgreSQLPgBouncerRow Sharding

Redis Cluster In-Memory Cache

< 2ms Latency

Distributed Redis cluster for calendar availability matrices, dynamic night pricing calculations (₹28,499), rate limiting, and guest reservation locks.

Redis 7Cluster ShardingPub/Sub Eviction

Object Storage (S3 / R2)

99.999999999% Durability

High-durability geo-replicated object storage for listing assets, host avatars, and 43 original RAW property photos with automated lifecycle archiving.

AWS S3Cloudflare R2Multipart Upload

CDC Event Streaming

Zero Read Amplification

Debezium captures PostgreSQL WAL events and streams mutations to Apache Kafka to synchronize search engines and cache invalidation instantaneously.

KafkaDebezium CDCEvent-Driven

